Assignment 1

Name: _____ Student ID: _____ Marks:_____

1. Determine u and v such that

$$\mathrm{e}^{z^2} = u(x, y) + \mathrm{i}v(x, y).$$

Show that u and v satisfy the Cauchy–Riemann equations for all (x, y). [5]

2. Given a function $f(z) = \mathrm{i}z + |z|$. Determine the points, if any, where $f(z)$ is differentiable.

[5]

3. Show that $\mathrm{Re}(\mathrm{i}z^2) = -2\mathrm{Re}(z)\mathrm{Im}(z)$. [5]

4. Determine where $f'(z)$ exists and find its value for [5]

(a) $f(z) = \dfrac{1}{z^2}$.

(b) $f(z) = z\mathrm{Im}(z)$.